Data Distributions

Data → A piece of information.

Data Distribution → A distribution explaining the data.

Types of Data Distributions :-

1. Discrete Distribution ⇝ Bar Chart
2. Continuous Distribution ⇝ Density Chart

What is discrete & continuous data ?

Discrete Data → Countable Values [Ex- days of a week, etc.]

Continuous Data → Uncountable Values [Ex- Height, temperature, etc.]

* Discrete data is categorical
  - strings
  - Numbers (countable)

* Continuous data is always numeric (uncountable nos.)

* Definitions:-

Random Variable (X) :- Outcome of a random experiment

Ex- Coin Toss

Let X = No. of heads obtained

So, P(X=H) ⇝ Prob. of getting a head on tossing a coin

Two types of Random Variable :-

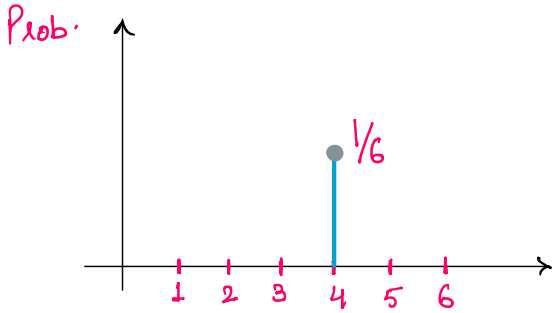
1. Discrete Random Variable → Ex- No. of students in a class
2. Continuous Random Variable → Ex - Height of a person

Probability Mass Function [PMF] :-

Point Mass → Prob. of something at a particular value

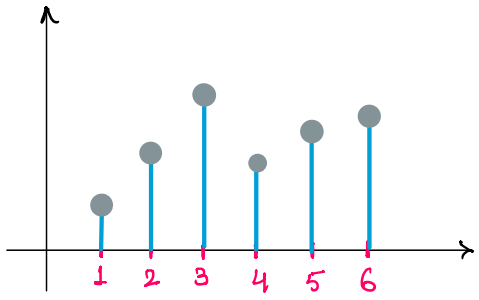
Collection of all point masses creates a PMF

Ex- Rolling a die & getting a 4



POINT MASS

Ex- Rolling a die multiple times



PMF

Cumulative Distribution Function [CDF] :-

Prob. that a random variable is less than or equal to a given value.

Formula →

$$F(x) = P(X \leq x)$$

Ques. What is the prob. of getting a 3 or less than 3 on rolling a die?

$$P(X \leq 3) = P(X=1) + P(X=2) + P(X=3)$$

$$= \frac{1}{6} + \frac{1}{6} + \frac{1}{6}$$

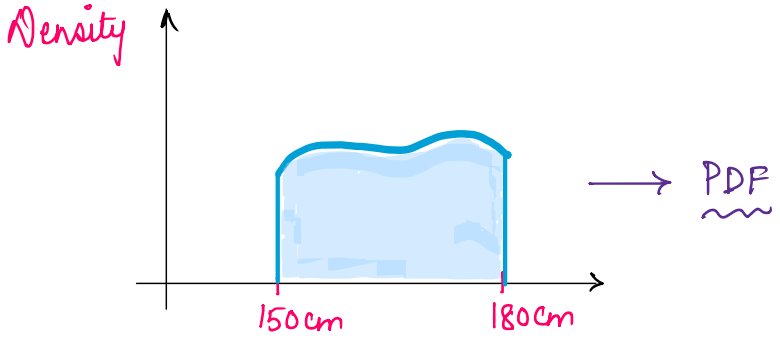
$$= \frac{3}{6} = \frac{1}{2} = 0.5 \rightsquigarrow \text{CDF}$$

Probability Density Function [PDF] :-

- It is used in case of continuous random variable
- It tells us that how a probability is distributed over a

- It tells us that how a probability is distributed over a range of values.

Ex- Distribution of heights of people



* Types of Discrete Distributions :-

① Bernoulli's Distribution :-

It gives the distribution of a single experiment with only two possible outcomes.

Let Random Variable (X) be

$$X = \begin{cases} 1, & \text{success} \\ 0, & \text{failure} \end{cases}$$

∴ Prob (success) = p

Prob (failure) = $1-p$

Formula →

$$P(X=x) = p^x \cdot (1-p)^{1-x}$$

where, $x \in \{0, 1\}$

p = Prob. of success $(x=1)$

$1-p$ = " " failure $(x=0)$

② Binomial Distribution :-

It gives the distribution of multiple experiments with only two possible outcomes.

Ex- Tossing a coin 5 times, etc.

Formula →

$$P(X=x) = {}^{n}C_{x} \cdot p^{x} \cdot (1-p)^{n-x}$$

where,

n = no. of trials

x = event/outcome

Permutation :- Arranging items by keeping an order.

Ex- ABC (Students) — □ □ (Chairs)

Permutation tells us in how many ways the three students can sit on the two chairs

A B
B A
A C
C A
B C
C B
} 6 ways

$$\text{Permutation} = {}^{n}P_{r}$$

$$= \frac{n!}{(n-r)!} \qquad n=3,\ r=2$$

$$= \frac{3!}{(3-2)!} = \frac{3\times2\times1}{1}$$

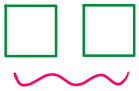
$$= 6$$

Combination :- Arranging items without considering the order.

Ex- ABC — □ □

Ex- ABC

Students



Chairs

AB
BC
AC
} 3 ways

Combination → ${}^{n}C_{r}$

$= \frac{n!}{r!\,(n-r)!}$

$n = 3$
$r = 2$

$= \frac{3!}{2!\,(3-2)!}$

$= \frac{3 \times 2 \times 1}{2 \times 1 \times 1}$

$= 3$